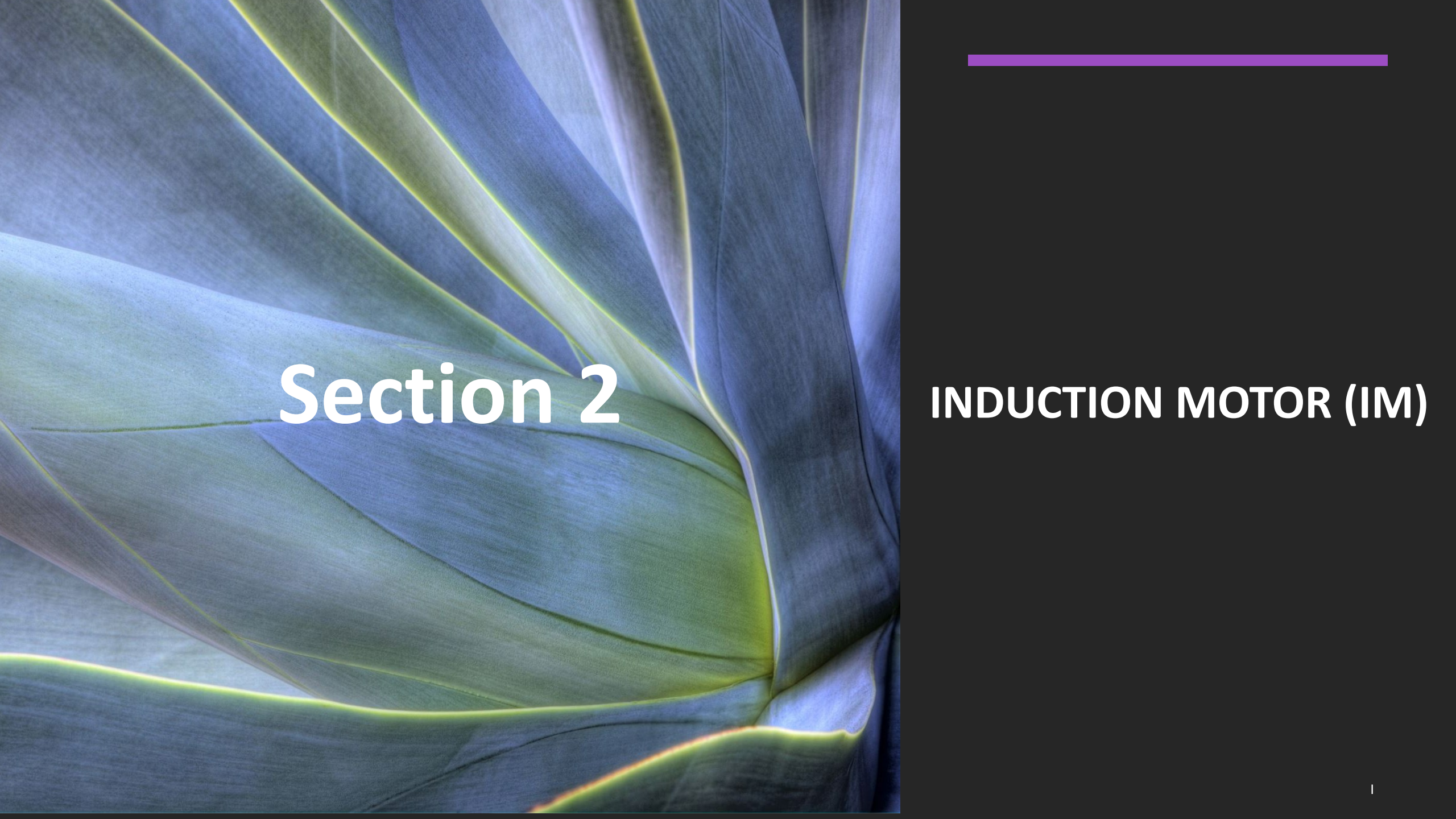




Section 2

INDUCTION MOTOR (IM)

Simulation Test 2

There are various methods for the speed control of an Induction Motor.

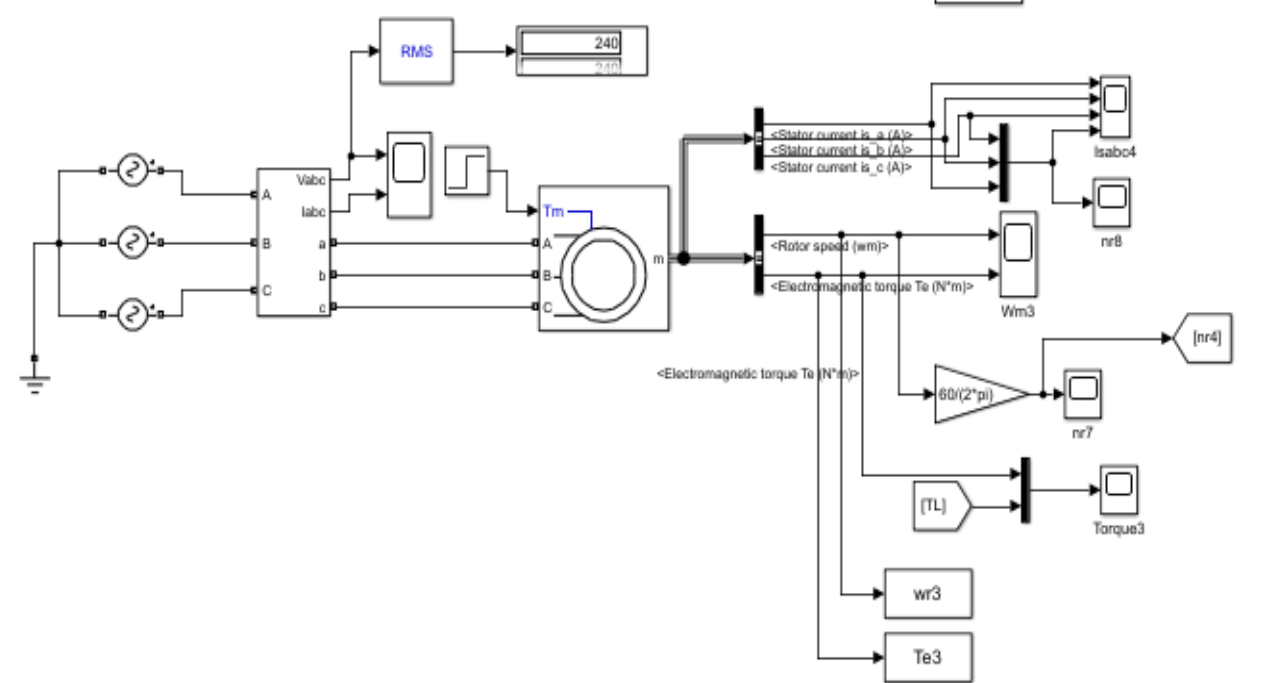
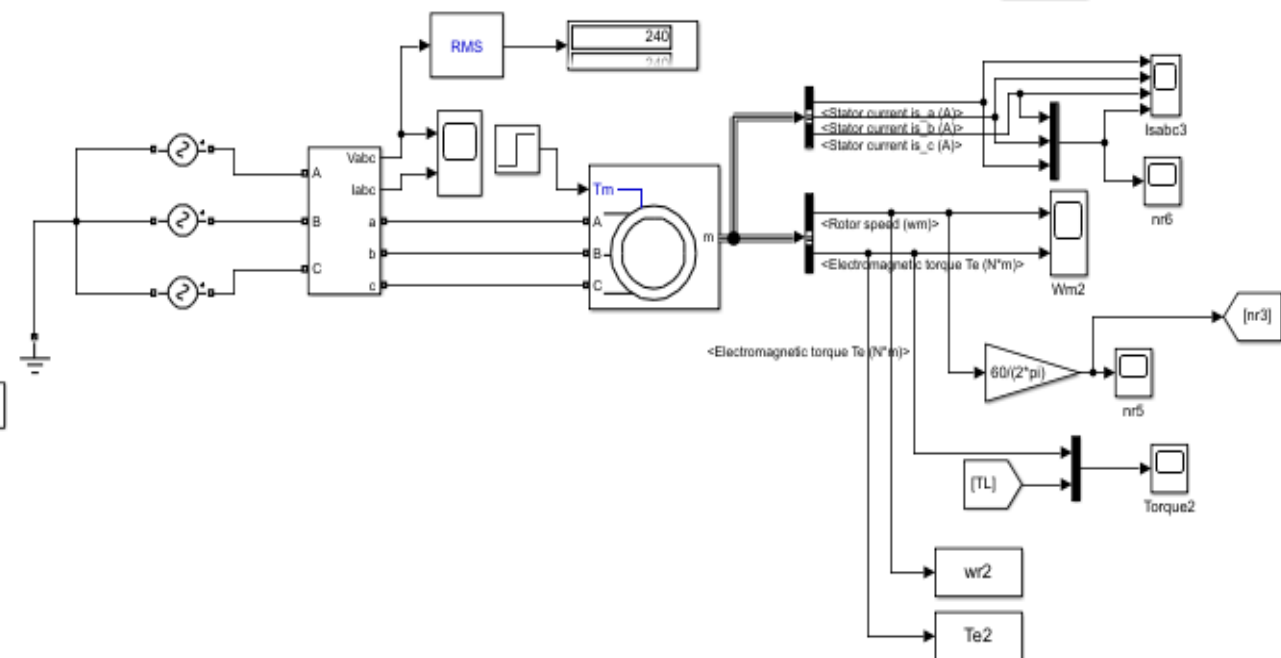
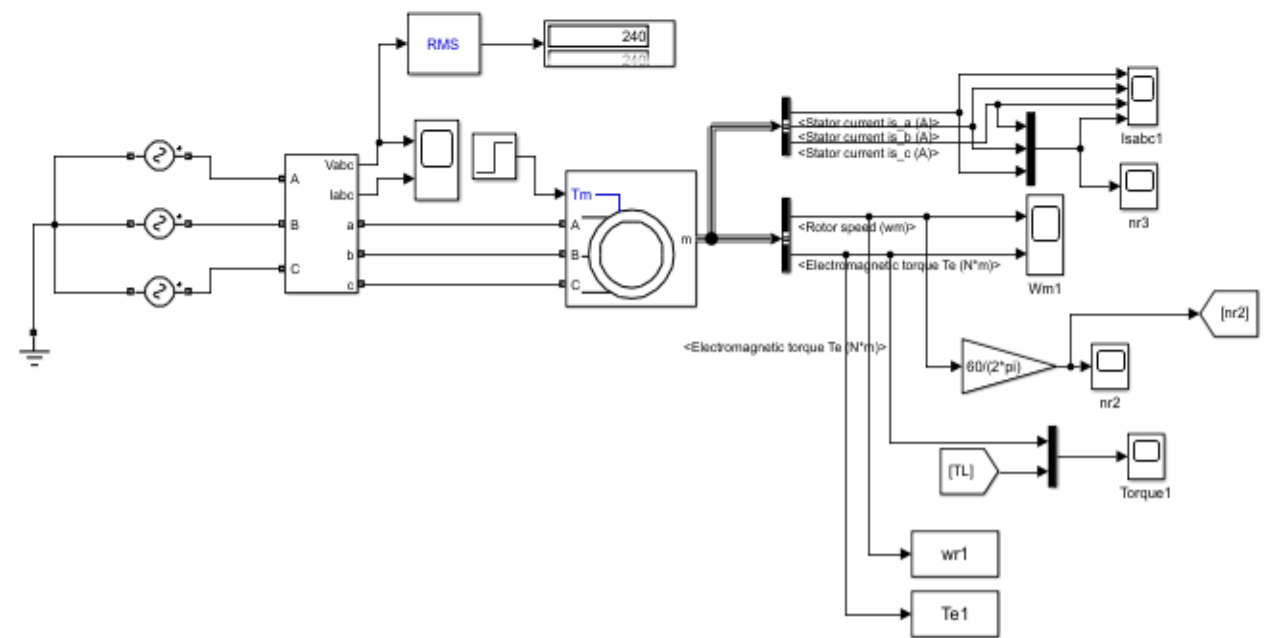
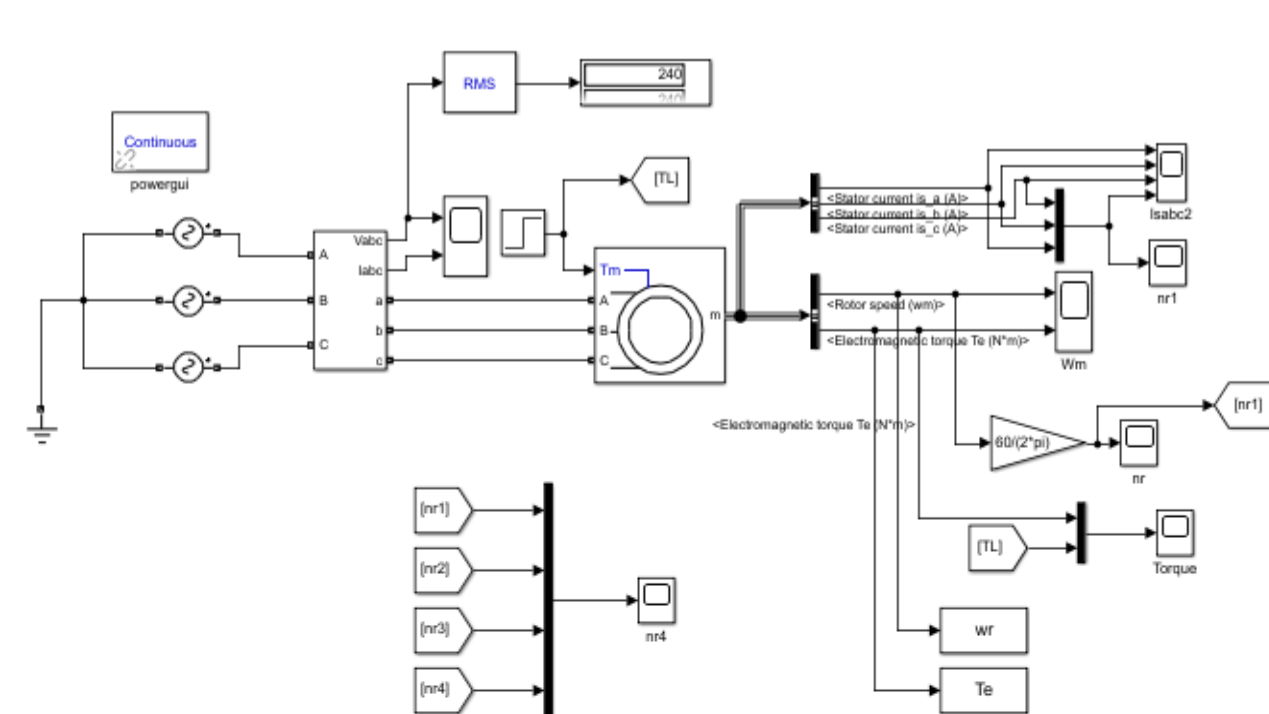
They are:

- Pole Changing
- Variable Supply Frequency
- Variable Supply Voltage

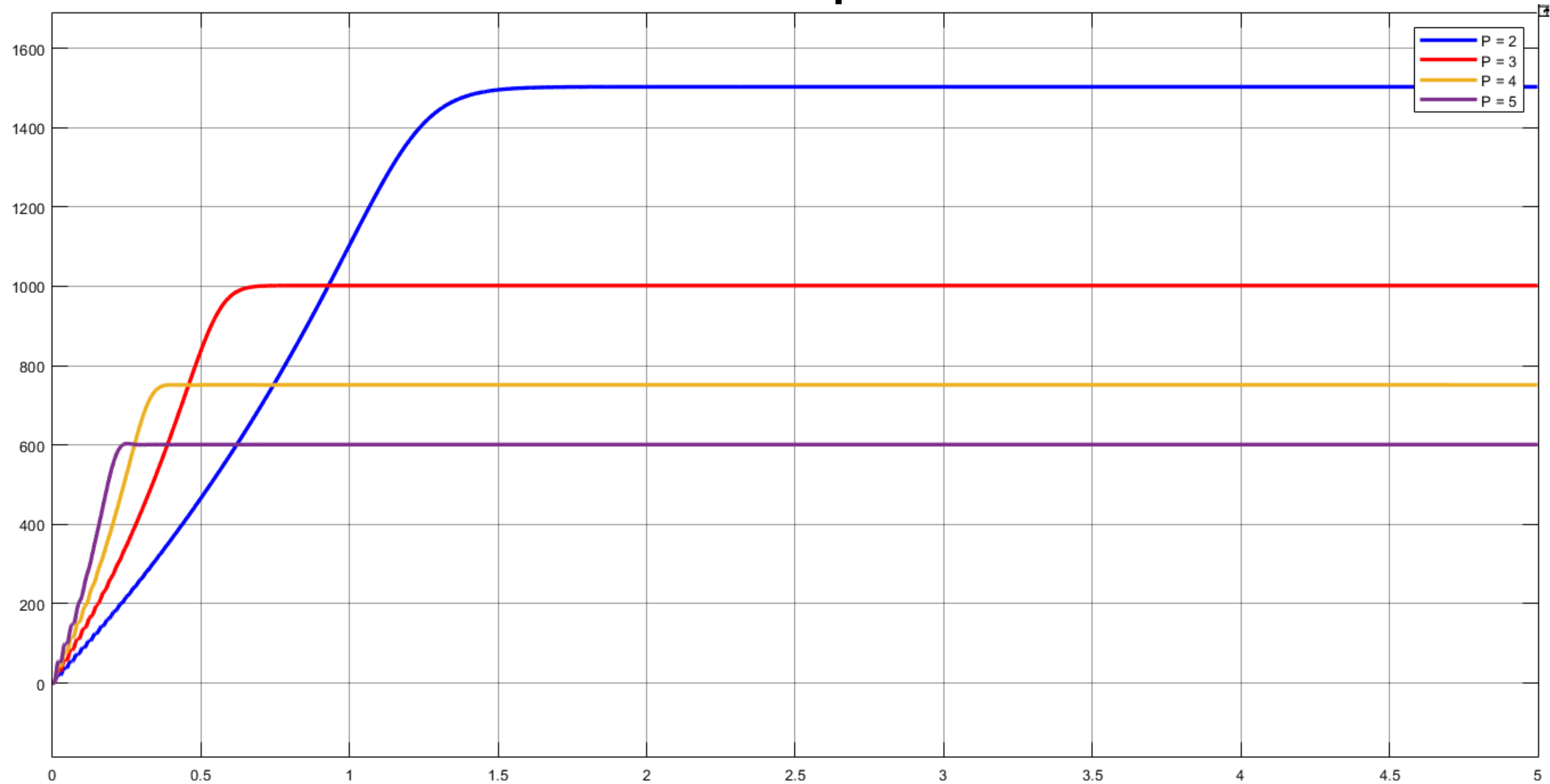
I. Poles Changing (Task I)

Plotting graph of Speed and Torque when the motor poles changed to 4, 6, 8, and 10.

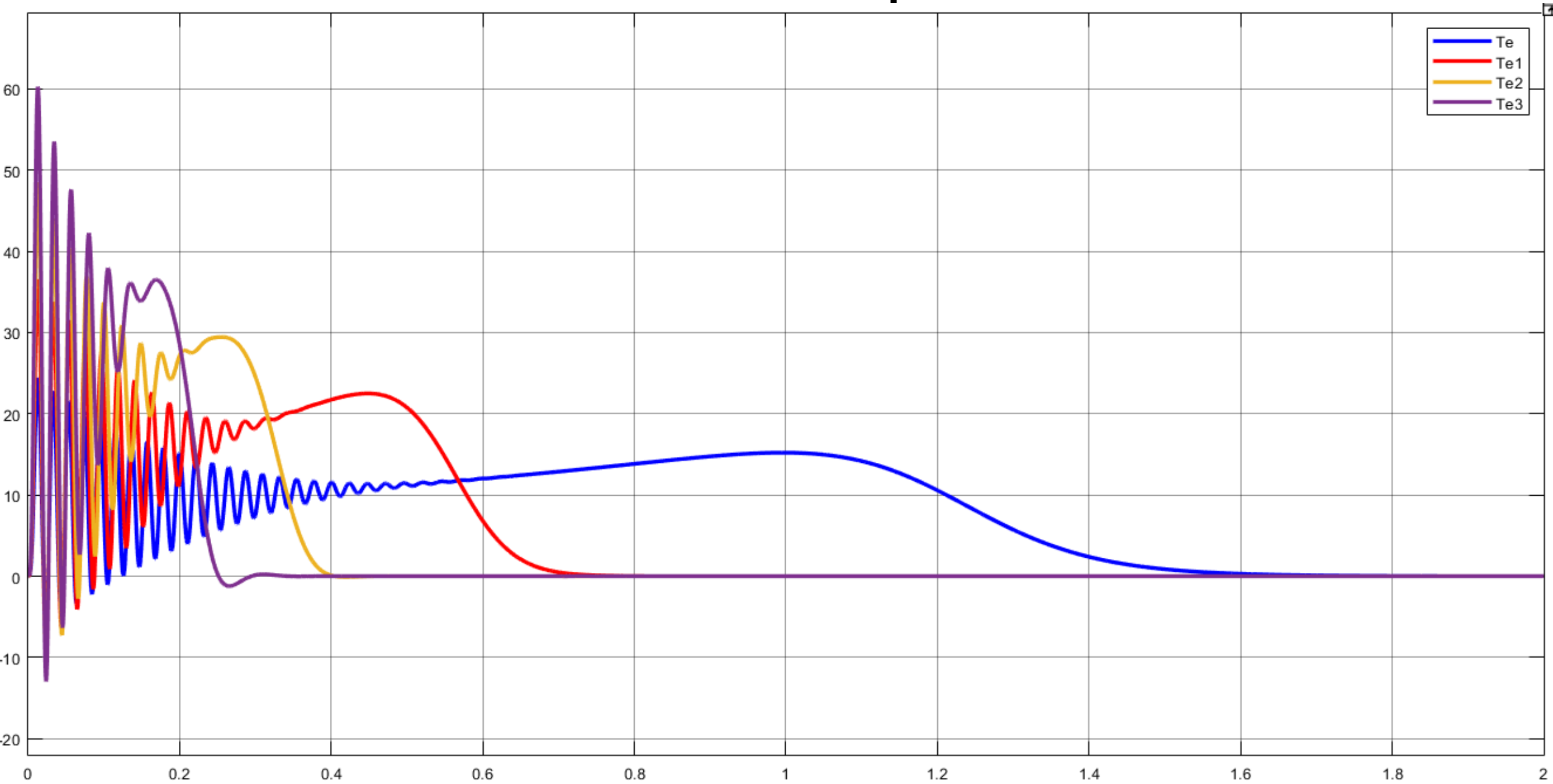
$$n_s = \frac{120f}{P}$$



Motor speed



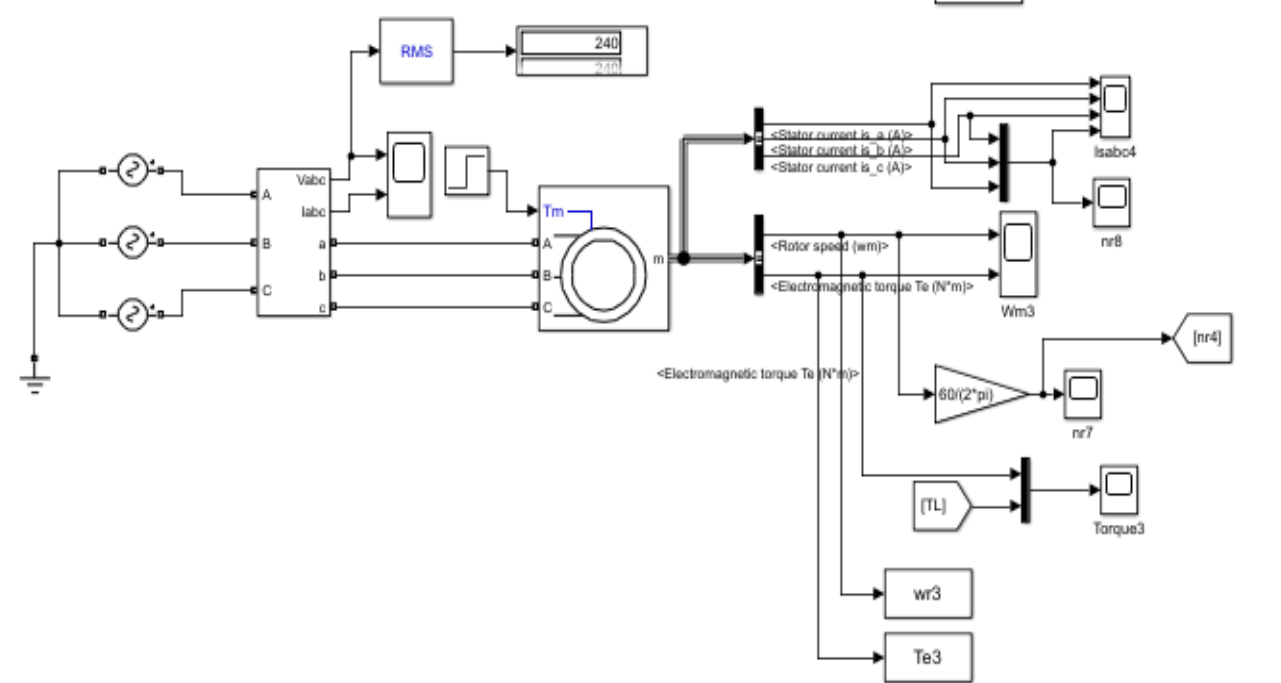
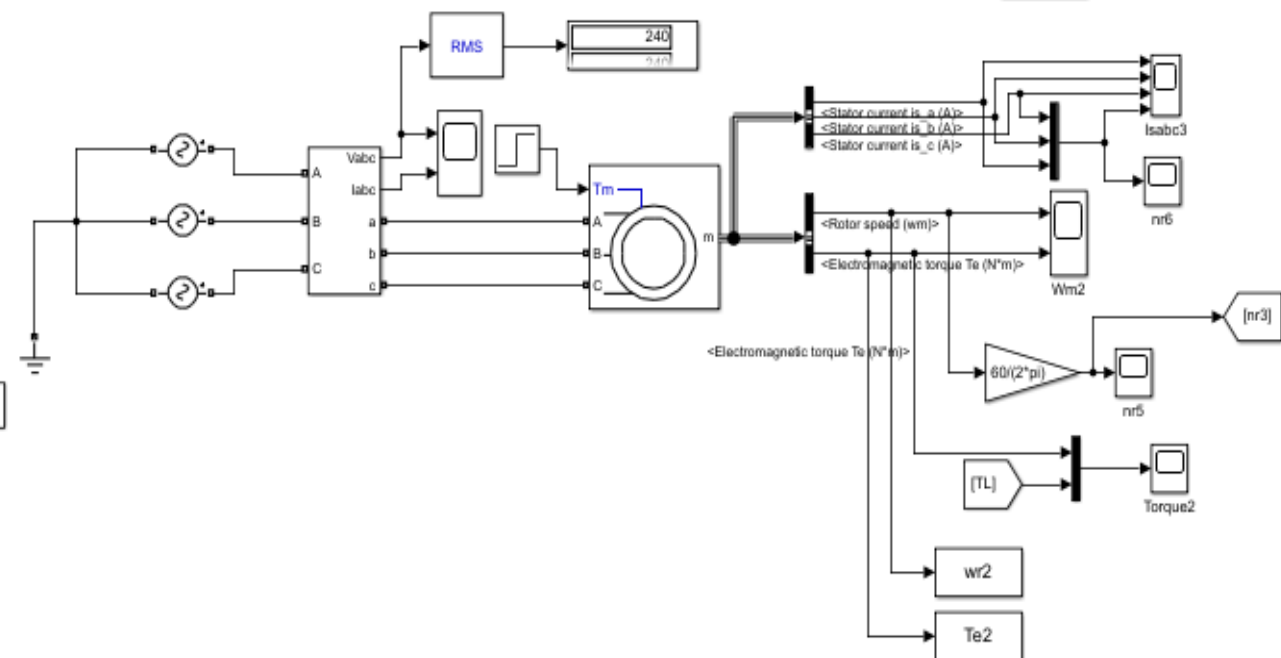
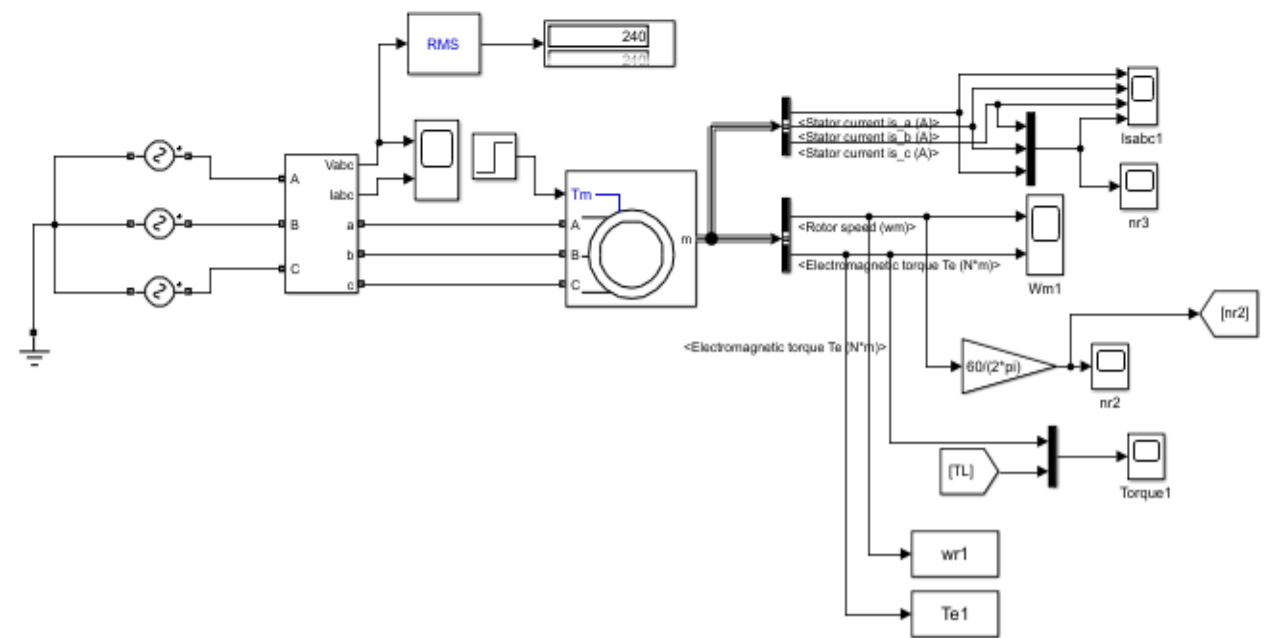
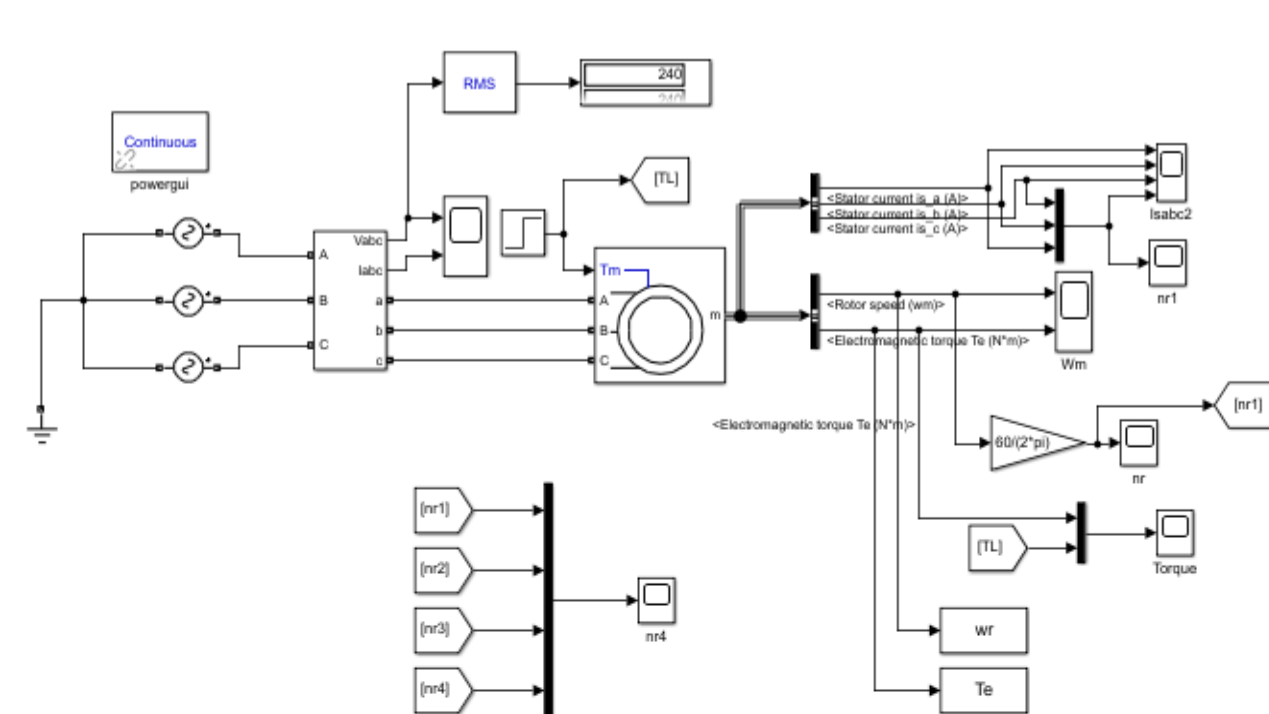
Motor torque



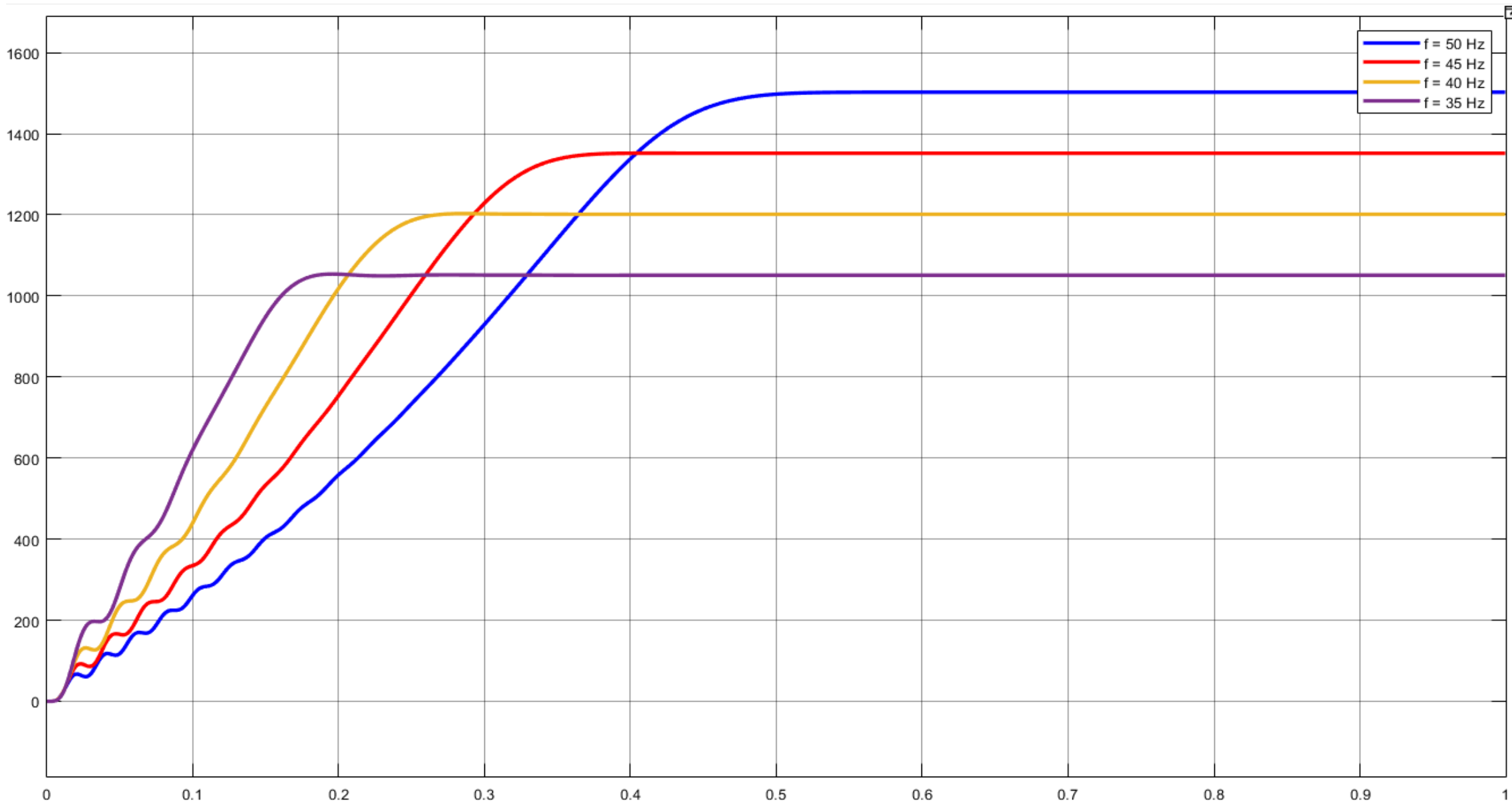
2. Frequency Changing (Task 2)

Plotting graph of Speed and Torque when the supply frequency changed to 50 Hz, 45Hz, 40Hz, and 35Hz.

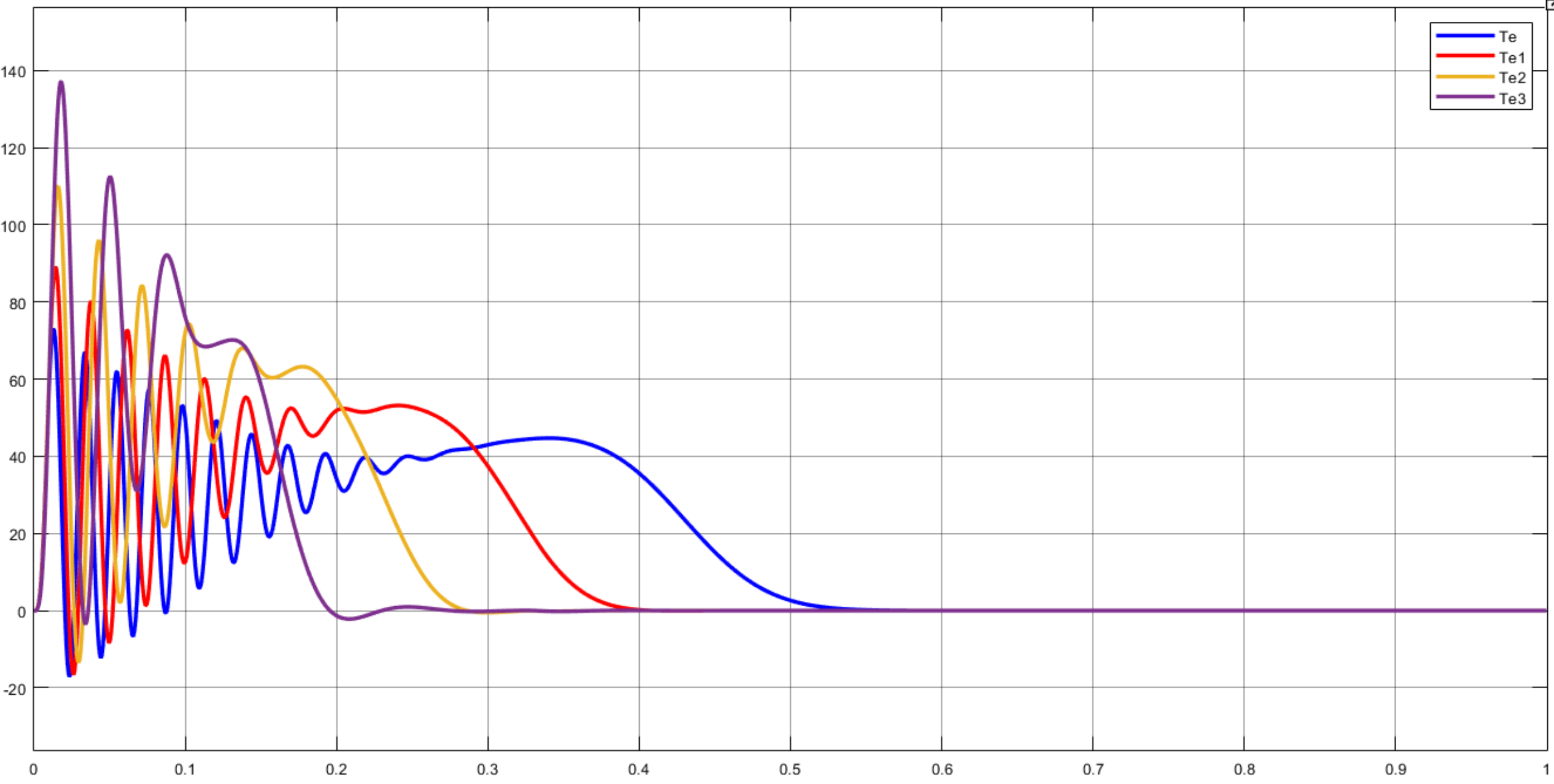
$$n_s = \frac{120f}{P}$$



Motor speed



Motor torque



3. Variable Supply Voltage (Task 3)

Plotting graph of speed and torque when the supply voltage (line-to-line) changed to 240V, 200V, 150V, and 100V.